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AIR PRE-COOLING

Abstract

A system includes at least one radiative cooling material, and at least one air intake. The at least one radiative cooling material is arranged upstream of the at least one air intake. Air moves from the radiative cooling material to the at least one air intake, such that the air is cooled using the radiative cooling material prior to entering the least one air intake. The radiative cooling material and the air intake may be arranged on a roof of a building, and the air intake may provide the cooled air to inside the building. The air intake may include any one or more of an air vent, an air condition, or an air cooled condenser. The radiative cooling material may include openings through which the air flows before entering the at least one air intake, where the air flow through the openings causes the air to be precooled.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/559,693, filed Feb. 29, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

INTRODUCTION

[0002] The present disclosure is directed to radiative cooling systems for cooling air and other fluids. More specifically, the present disclosure is related to precooling air that, for example, enters any suitable air intake (e.g., which may itself be an air intake system, or which may be a component of an air intake system), such as may be coupled to HVAC/R equipment, a building, or any other suitable structure. As used herein, air may be regarded as “precooled” if it is cooled prior to entering the air intake.

SUMMARY

[0003] Air intakes (e.g., roof or building intake vents) and air handling equipment (e.g., condensers, fluid coolers, any suitable HVAC equipment, and other air cooling devices or air exchange systems) are often located on the roofs of buildings. On certain days, particularly in summers and especially in hot climates, certain roofs (e.g., dark and white roofs) may be hotter than the ambient air (e.g., above the roof or around the building). Such hot roofs may heat up the air that travels or resides proximal to the air intake system (e.g., including an air intake and optionally air handling equipment). As a result, the air received at the air intake can be warmer than the ambient air. Thus, a cooling system (e.g., an HVAC system and/or any other suitable cooling system, which may include the aforementioned air handling equipment) coupled to the air intake may be forced to cool air that is warmer than the ambient air. Being forced to cool warmer air increases the associated energy requirements of the cooling system, and may cause the cooling system to output warmer air than it would if it received the intake air at the ambient temperature. Thus, that warmer intake air may reduce comfort or exacerbate cooling requirements of the internal building space.

[0004] In accordance with embodiments of the present disclosure, radiative cooling systems are provided for precooling air (i.e., cooling the air prior to the air enters an air intake) and other fluids that enter an air intake (e.g., of HVAC/R equipment, of a building, of any suitable air intake system, of any other suitable structure, or of any combination thereof). As mentioned, the air intake may be coupled to a cooling system, which may include any suitable air handling equipment (e.g., a condenser, evaporator, heat pump, or any other cooling device). In some embodiments, a system includes a roof, at least one radiative cooling material on the roof, and a fluid exchange system on the roof, where the fluid exchange system permits a working fluid (including, but not limited to air) to enter and exit the building under the roof, and the at least one radiative cooling material cools the working fluid before it enters the building. In some embodiments, the fluid exchange system is any of an air conditioner, condenser, vent, or other air conveyance system. In some embodiments, the fluid is air; in other embodiments, the fluid is a liquid including water or any suitable coolant. In some embodiments, radiative cooling materials are integrated with the air conveyance system or are otherwise configured to form a particular air flow path, such that air entering the conveyance system is more exposed to the cooling surface of the radiative cooling material, and therefore is cooled to a greater degree.

[0005] In accordance with embodiments of the present disclosure, a system includes at least one radiative cooling material, and at least one air intake. The at least one radiative cooling material is arranged upstream of the at least one air intake. Air moves from the radiative cooling material to

the at least one air intake, such that the air is cooled using the radiative cooling material prior to entering the least one air intake. The radiative cooling material and the air intake may be arranged on a roof of a building, and the air intake may provide the cooled air to inside the building. The air intake may include any one or more of an air vent, an air condition, or an air cooled condenser. The radiative cooling material may include openings through which the air flows before entering the at least one air intake, where the air flow through the openings causes the air to be precooled.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The above and other objects and advantages of the disclosure will be apparent upon consideration of the following detailed description, taken in conjunction with the accompanying drawings, in which like reference characters refer to like parts throughout, and in which:

[0007] FIG. 1 is an illustrative depiction of the air entering a condenser exceeding the ambient temperature in accordance with some embodiments of the present disclosure;

[0008] FIG. 2 is an illustrative depiction of an air cooled condenser with the air intake cooled by radiative cooling films in accordance with some embodiments of the present disclosure;

[0009] FIG. 3 is an illustrative depiction of a roof vent with the air intake cooled by a radiative cooling material in accordance with some embodiments of the present disclosure;

[0010] FIG. 4 is a first illustrative depiction of a roof vent with a baffled radiative cooling material for cooling and directing air intake to the vent in accordance with some embodiments of the present disclosure;

[0011] FIG. 5 is a second illustrative depiction of a roof vent with a baffled radiative cooling material for cooling and directing air intake to the vent in accordance with some embodiments of the present disclosure;

[0012] FIG. 6 is an illustrative depiction of a roof vent with a radiative cooling material, a thermal mass, and a wind screen for cooling and directing air intake to the vent in accordance with some embodiments of the present disclosure;

[0013] FIG. 7 is an illustrative depiction of a roof vent with an air-permeable radiative cooling material, a thermal mass, and a wind screen for cooling and directing air intake to the vent in accordance with some embodiments of the present disclosure;

[0014] FIGS. 8A-8B are first illustrative depictions of a roof vent with a perforated radiative material arranged over a support grid for cooling and directing air intake to the vent in accordance with some embodiments of the present disclosure;

[0015] FIGS. 9A-9B are second illustrative depictions of a roof vent with a perforated radiative material arranged over a support grid for cooling and directing air intake to the vent in accordance with some embodiments of the present disclosure;

[0016] FIG. 10 is an illustrative depiction of a roof vent with a radiative cooling material and a heat exchange fin for cooling and directing air intake to the vent in accordance with some embodiments of the present disclosure;

[0017] FIG. 11 is an illustrative depiction of a roof with a radiative cooling material, a wind screen, and a perimeter wall for cooling and directing air intake to the vent in accordance with some embodiments of the present disclosure;

[0018] FIG. 12 is a first illustrative depiction of a radiative cooling surface coupled to a thermal mass through a heat conduction pathway in accordance with some embodiments of the present disclosure;

[0019] FIG. 13 is a second illustrative depiction of a radiative cooling surface coupled to a thermal mass through a heat conduction pathway in accordance with some embodiments of the present disclosure;

[0020] FIG. **14** is an illustrative radiative cooling system for cooling air flow into a building in accordance with some embodiments of the present disclosure;

[0021] FIG. **15** is an illustrative radiative cooling system with an internal barrier for releasing warm air and directing cool air to a roof vent in accordance with some embodiments of the present disclosure;

[0022] FIG. **16** is illustrative cross-section and isometric views of a radiative cooling system with an angled plenum and coupling to a rooftop vent in accordance with some embodiments of the present disclosure; and

[0023] FIG. **17** shows a flowchart of an illustrative method for air precooling, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0024] FIG. **1** shows illustrative temperature measurements **100** recorded at a building roof in accordance with some embodiments of the present disclosure. The top panel of FIG. **1** includes illustrative data that show temperature **110** measured under a condenser (e.g., of an air handling system of the building), which is denoted as “air temperature entering an air cooled condenser,” and temperature **120** measured with an aspirated radiation shield at a distance of 1 m above the roof, which is denoted as “ambient air temperature measured with a radiation shield.” The bottom panel of FIG. **1** shows the temperature difference **130** (e.g., temperature 110 minus temperature 120). On average, the condenser sees an inlet air temperature that is 3 to 5 degrees Fahrenheit warmer than the true ambient air temperature due to heating of the roof. Because of how the sun heats the roof, air into the condenser is warmed by the hot roof, increasing the resulting cooling load of the condenser. This additional heat represents an energy and/or performance penalty to the HVAC equipment and adds to the total cooling load of any building with a hot roof (e.g., which may characterize any building for which the temperature of air entering the building via a roof intake system is greater than the temperature of ambient air).

[0025] To overcome that energy and/or performance penalty, in accordance with some embodiments of the present disclosure, radiative cooling materials, structures, systems, and methods are disclosed for precooling air (e.g., at rooftop interfaces), such that air-handling equipment (e.g., on the roof) can operate more efficiently. In some embodiments, for every 1 degree Fahrenheit reduction in the intake air temperature to an air cooled condenser, the connected refrigeration or AC system could see a 1% improvement in its efficiency. In some embodiments, the connected air system could realize even greater benefits if the radiative cooling system extends the number of hours in which outdoor economization (e.g., the intake of outdoor air into the air handling system) can occur.

[0026] In accordance with some embodiments of the present disclosure, radiative cooling materials, films, panels, or other structures are applied to cool rooftop air. In some embodiments, various arrangements of the radiative cooling material are disclosed to improve a cooling efficiency of the system. In some embodiments, buildings have a high amount of roof insulation, such that putting radiative cooling material directly on the roof won't be very effective for cooling. Therefore, structures are disclosed to directly cool air before it enters the building, vent, condenser, or other air handling structure.

[0027] FIG. **2** shows an illustrative structure **200** that includes a radiative cooling material **201** (e.g., a radiative cooling film) to precool air into a condenser or fluid cooler in accordance with some embodiments of the present disclosure. Air is suctioned around or through the radiative cooling surface (which may have holes, slits or other perforations in it), as indicated by air intake flow paths **202**. Due to interfacing with the radiative cooling material **201**, the air is cooled before it enters the air handling system **203**. In some embodiments, as shown in FIG. **2**, the air intake system is an air cooled condenser with an air intake **202** and an air exhaust **204**, although any suitable air intake equipment could be used. In some embodiments, the radiative cooling material is arranged above the roof (e.g., there is a gap between the cooling material and the rooftop). Intake

air flows through the gap and is cooled before entering the air cooled condenser. In some embodiments, a seal **205** is arranged such that warm air outside the air intake plenum cannot enter the air cavity and mix with the intake air, and such that air cooled by the radiative cooling films cannot exit the air cavity. Structure **200** is arranged on top of roof **206**, under which may be the inside of any suitable building.

[0028] FIG. **3** shows an illustrative structure **300** that includes a radiative cooling material **301** directly applied to the roof **303** around an air intake vent **302**, and to the air intake vent **302** itself, in accordance with some embodiments of the present disclosure. Based on the geometry of the vent and the natural flow path driving air intake **304**, the intake air to the vent **302** is cooled by virtue of traveling over the radiative cooling material **301**.

[0029] As shown at least in FIGS. **3-5**, and as may be applied to any embodiments of the subject matter of this disclosure, a radiative cooling material may be applied to the underside of a roof vent so that it is not directly exposed to sunlight. As a result, the radiative cooling material can maintain a lower temperature and thereby provide better cooling performance.

[0030] FIG. **4** shows an illustrative structure **400** that includes a radiative cooling surface **401** (e.g., a panel, scaffold, or other structure to which the radiative cooling material is applied, e.g., as a film) arranged above the roof **410** and with an air gap **403** between the cooling surface **401** and the roof **410**, in accordance with some embodiments of the present disclosure. Based on the geometry of the radiative cooling surface **401**, air is oriented to a trough surrounding the roof vent **404**. The particular geometry of the radiative cooling material, which includes a trough, is realized by a corresponding arrangement of the structure on which the radiative cooling material is applied. Support for the structure **405** extends from the top of the roof to the base of the structure, as shown. Air in the trough (and air contacting the baffled cooling material oriented toward the trough) is cooled by the radiative cooling material and thus made more dense. As a result, the colder air is trapped or held around the air vent because it is more dense than the surrounding air. Moreover, the air of air intake **406** is cooled before entering roof vent **404**.

[0031] In some embodiments, structure **400** also includes a radiative cooling material **402** applied to the top and/or bottom sides of roof vent **404**. Radiative cooling material **402** may be the same material as is applied over radiative cooling structure **401**. Radiative cooling material **402** can further cool the air of air intake **406**, and radiative cooling material **402** can also cool the roof vent **404**. Having a cooler roof vent would further contribute to precooling of air before it enters a condenser or other components of an air handling system.

[0032] FIG. **5** shows an illustrative structure **500** that includes an air gap **503** between the roof **510** and the radiative cooling structure **501**, where the air gap is **503** configured to provide an air cavity that provides cooled air at air intake **506** to roof vent **504**, in accordance with some embodiments of the present disclosure. In some embodiments, radiative cooling structure **501** includes a flexible substrate over which the radiative cooling material is applied. The flexible substrate and the radiative cooling material can be arranged (e.g., using tension or any other suitable force) around the rooftop intake vent **504** to direct cooled air toward the roof vent intake, as per air intake **506**. As mentioned above, the flexible substrate may be configured to seal off the air cavity from air that is above (with respect to the rooftop) the radiative cooling structure **501**. In some embodiments, the radiative cooling material included in radiative cooling structure **501** may have holes, slits, perforations, or other permeability features such that air can flow through the radiative cooling structure **501**. In some embodiments, water (e.g., from precipitation or condensation) can be drained from the surface and optionally collected (e.g., similar to the arrangements shown in FIGS. **13-14**).

[0033] In some embodiments, structure **500** also includes a radiative cooling material **502** applied to the bottom side of roof vent **504**. Radiative cooling material **502** may be the same material as is applied over radiative cooling structure **501**. Radiative cooling material **502** can further cool the air of air cavity **503** and of air intake **506**. Any roof vent, including all of those described in this

disclosure, may have a radiative cooling material applied to its top side and/or its bottom side. For brevity, such top-side and/or bottom-side radiative cooling materials may not be explicitly mentioned in connection with other embodiments of this disclosure, despite those top-side and/or bottom-side radiative cooling materials being included or considered in connection with those other structures.

[0034] FIG. 6 shows an illustrative structure **600** that includes a radiative cooling material **601** arranged below a wind screen **603** and above (e.g., in direct contact with) a thermal mass **605**, all of which is arranged over roof **610**, in accordance with some embodiments of the present disclosure. In some embodiments, the wind screen **603** is made of a mesh or an IR transparent material. In some embodiments, the wind screen **603** is transparent to visible light. The wind screen **603** may prevent convection from warming up the surface of radiative cooling material **601**, and may also keep this surface clean. A cooling power of radiative cooling material **601** may be made worse by the accumulation of dirt, debris, or other materials (e.g., the radiative cooling material is most effective when it is clean). Thus, the wind screen **603** may improve the performance of radiative cooling material **601** due to preventing convective warming and further due to keeping its surface clean. In some embodiments, the wind screen **603** has snaps, hooks or other attachments that can let the wind screen **603** be readily applied to and/or removed from the radiative cooling material **601**. In some embodiments, the wind screen **603** may be removed or replaced every cooling season or at any other suitable frequency.

[0035] FIG. 6 additionally shows a thermal mass **605** under the radiative cooling material **601**. As further described below, the thermal mass **605** may include a material that maintains a low temperature (due to being cooled by the radiative cooling panel) and thereby provides an additional cool surface that is able to draw heat away from the rooftop air (e.g., that enters roof vent **604**).

[0036] FIG. 7 shows an illustrative structure **700** that includes an air-permeable (e.g., based on being perforated or stamped) radiative cooling material **701** arranged below a windscreen **703** on a first side of roof vent **704** and above (e.g., in direct contact with) a thermal mass **705** on a second side of roof vent **704**, in accordance with some embodiments of the present disclosure. The air-permeable radiative cooling material **701** may have an improved cooling power as compared to an air-impermeable (e.g., based on being unperforated or unstamped) radiative cooling material due to having more surface area that is exposed to air, and further due to altering convection along the surface of the material. As mentioned, cooled air may become more dense than surrounding warmer air. That denser, cooled air may stay inside the air gap between roof **710** and cooling structure **700** (e.g., which may be referred to as an air intake cavity) even though the air gap may not be sealed (e.g., due to the stamping or perforation in the air-permeable radiative cooling material **701**). Even so, to limit the permeability of the air gap, the lateral edge of the air gap may be sealed against the top of roof **710**. As a result, as shown, there may be no direct flow path (or limited direct flow paths) into the air gap besides those paths through the air-permeable radiative cooling material **701**.

[0037] For the reasons given above, any radiative cooling material described in connection with embodiments of this disclosure may be made permeable to air (e.g., based on a perforation, stamping, or other suitable process). A degree of the permeability may be configured to maximize a radiative cooling power of the corresponding air-permeable radiative cooling material. For example, the permeability may be configured to increase surface area without causing significant mixing of cool air (e.g., under, or in contact with, the radiative cooling material) and warmer surrounding air (e.g., that is not affected by the radiative cooling material). At the same time, any air-permeable radiative cooling material described in connection with embodiments of this disclosure may be made impermeable to air and still serve radiative cooling applications in connection with a corresponding structure.

[0038] FIG. 8A shows a top-down view of illustrative structure **800**, including a roof vent **804**, a support grid **808** (e.g., where the support grid holds up a support to which the radiative cooling

material **801** is applied), and a radiative cooling material **801** (as further described in connection with FIG. **8B**) over a roof **810**, in accordance with some embodiments of the present disclosure. In the top-down view of FIG. **8A**, the radiative cooling material **801**, despite being present, is not explicitly shown to better depict the underlying support grid **808**. As shown, a curb or shim **809** may be glued to the roof **810** and surrounding the support grid **808**. Illustrative and non-limiting sizes associated with structure **800** may be 10 ft×10 ft for the roof vent **804** and 50 ft×50 ft for roof **810** (or, at least 50 ft×50 ft for roof **810**, where the curb or shim **809** surrounding the support grid **808** occupies 50 ft×50 ft, and support grid **808** occupies a comparable area). These illustrative sizes may scale with the size/number of roof vents and/or the size/shape of the roof.

[0039] FIG. **8B** shows a side view of illustrative structure **800**, in connection with the top view of FIG. **8A**, in accordance with some embodiments of the present disclosure. Radiative cooling material **801** is permeable to air (e.g., due to being perforated, as annotated, stamped, or otherwise made permeable) and is arranged on top of support grid **808** to create an air gap between the top of the roof **810** and the bottom of radiative cooling material **801**. In some embodiments, the air gap is 1 foot in height; in other embodiments, other suitable air gap heights are also considered. In some embodiments, radiative cooling material **801** (e.g., based on being adhered to support grid **808**) is connected to the roof **810** through connection to curb or shim **809**, as shown.

[0040] Similar to FIG. **8A**, FIG. **9A** shows a top-down view of illustrative structure **900**, which includes a circular support grid over a roof, in accordance with some embodiments of the present disclosure. Similar to FIG. **8B**, FIG. **9B** shows a side view of illustrative structure **900**, in accordance with some embodiments of the present disclosure. FIG. **9B** depicts how a radiative cooling material may be arranged on the roof with an air gap and may be connected to the roof through the curb or shim. Other than the shape of the support grid **908** and the corresponding shape of radiative cooling material **901**, structure **900** may correspond to structure **800**. That is, the descriptions of structure **800** may apply to structure **900**, and are not repeated for brevity.

[0041] FIG. **10** shows illustrative structures **1000** and **1050**, each of which includes a fin **1003** with a corresponding radiative cooling material and a roof vent **1004**, arranged over a roof **1010**, in accordance with some embodiments of the present disclosure. In some embodiments, the fin **1003** provides extra surface area with which to cool air that enters roof vent **1004** and thereby improves radiative cooling powers of the structures **1000** and **1050**. For example, as shown, the fin **1003** may be arranged within at least a portion of the air cavity providing intake air to the roof vent **1004** (or any other suitable air handling equipment). In some embodiments, including as shown in structure **1000**, the fin **1003** is arranged beneath a flat surface and the radiative cooling material **1001** is arranged above the flat surface; thus, the radiative cooling material **1001** cools the fin **1003** through the flat surface. In some embodiments, including as shown in structure **1050**, the radiative cooling material **1051** (as depicted by the gray curve positioned over the black curve) is arranged in direct contact with the fin **1003** (as depicted by the black curve positioned under the gray curve). Thus, the radiative cooling material **1051** directly cools the fin **1003** through conduction.

[0042] FIG. **11** shows illustrative structure **1100**, including a radiative cooling material **1101**, roof vent **1104**, perimeter walls **1108** surrounding the radiative cooling material **1101** and the roof vent **1104**, and wind screen **1103** arranged over roof vent **1104**, all of which is arranged over roof **1110**, in accordance with some embodiments of the present disclosure. That is, radiative cooling material **1101** is applied to roof **1110**, enclosed within perimeter walls **1108**, and arranged under wind screen **1103**. In some embodiments, the perimeter walls **1108** and the wind screen **1103** both serve to reduce convective heating of the radiative cooling material **1101** and to keep the radiative cooling material **1101** clean. As shown, the radiative cooling material may be directly applied to roof **1110** without an air gap between the roof **1110** and the radiative cooling material **1101** (e.g., similar to the arrangement shown in FIG. **3**). In other embodiments, the radiative cooling material **1110** may be arranged with an air gap between the roof **1110** and the radiative cooling material **1110** (e.g., similar to any of the arrangements shown in FIGS. **4-10**).

[0043] FIG. **12** shows an illustrative structure **1200** that includes a radiative cooling material **1201** that is applied over a thermal conduction pathway **1203**, in accordance with some embodiments of the present disclosure. The thermal conduction pathway **1203** is thermally coupled to thermal mass **1205** (e.g., a brick or ballast, as annotated, or any other suitable thermal mass). The thermal mass **1205** provides thermal energy storage of a cooling power of radiative cooling material **1201**. Thermal conduction path **1203** thermally couples radiative cooling material **1201** the radiative cooling surface to the thermal mass. Thus, the radiative cooling material **1201** may cool the thermal mass **1205** and the cool thermal mass may precool air entering an air handling system or any other suitable structure or equipment.

[0044] FIG. **13** shows illustrative structures **1300** and **1350**, each of which represents an additional arrangement (compared to structure **1200**) for thermally coupling a radiative cooling material to a thermal mass, in accordance with some embodiments of the present disclosure. Structure **1300** includes a curved and concave-down radiative cooling material **1301**, which is applied over a correspondingly-shaped thermal conduction pathway **1303**. Thermal conduction pathway **1303** is thermally coupled to thermal mass **1305**. Structure **1350** includes a slanted and downward-sloping down radiative cooling material **1351**, which is applied over a correspondingly-shaped thermal conduction pathway **1353**. Thermal conduction pathway **1353** is thermally coupled to thermal mass **1305**. Structure **1300** and structure **1350** may be shaped (e.g., with concave-down or downward-sloping orientation) to prevent dirt from accumulating on the surface of the corresponding radiative cooling material. Structure **1300** and structure **1350** may also be shaped (e.g., with concave-down or downward-sloping orientation) for water collection, as further described below.

[0045] In some embodiments, structure **1300** and structure **1350** each include a water collector **1307**. For example, as shown, the water collector **1307** may be arranged at the edge of radiative cooling material **1301** or at the edge of radiative cooling material **1351**, such that the shape of the radiative cooling material directs gravity-driven water flows into water collector **1307**. As a result, condensation, dew, rain water, or other moisture that collects on radiative cooling material **1301** or on radiative cooling material **1351** may be collected. In some embodiments, the water collector **1307** may be integrated with a potable water system, such that the collected water can serve as drinking water or be used in other suitable tasks (e.g., cooling, irrigation, manufacturing, or any other suitable task).

[0046] FIG. **14** shows an illustrative structure **1400** including a radiative cooling material **1401** thermally coupled to an air intake system (e.g., including roof vent **1404**, fan **1406**, air cavity **1405**, and air inlet **1409**, and enclosed by perimeter walls **1408** and base material **1407**) on a roof **1410**, in accordance with some embodiments of the present disclosure. Structure **1400** may be used to control and precool air flowing into any suitable building. In some embodiments, including as shown in FIG. **14**, radiative cooling material **1401** is air permeable (e.g., based on being perforated, stamped, slitted, or otherwise having openings), sealed to perimeter walls **1408**, and further sealed to a perimeter of roof vent **1404**. Thus, all (or at least most) air residing in air cavity **1405** and entering air inlet **1409** (e.g., as indicated by the arrows depicting air flows) passes through the air-permeable radiative cooling material **1401**. Perimeter walls **1408** may be mechanically coupled to base material **1407** that is arranged above roof **1410** such that there is an air gap between the roof **1410** and base material **1407** (e.g., the gap being the same height as the support structures holding up the base material **1407**). The base material **1407** separates the air gap directly above roof **1410** from air cavity **1405**, which holds air that flows into the building via air inlet **1409**. As shown, the roof **1410** has an opening (which may, e.g., be a vent or other controllable opening) through which air may be pushed by fan **1406** (or otherwise driven, e.g., via convective forces) into the building. The air cavity **1405** is coupled to the opening of air inlet **1409** and holds precooled air, e.g., that may be cooled upon being pulled through an opening in radiative cooling material **1401** and upon otherwise interfacing with a bottom side of radiative cooling material **1401**. Fan **1406** directs this precooled air into the building (e.g., where it may enter an air handling, air conditioning, or other

suitable air exchange system).

[0047] As shown by the inset of FIG. **14**, in some embodiments, radiative cooling material **1401** is angled. For example, radiative cooling material **1401** may be shaped in a downward- sloping (with respect to the middle) manner, as shown by radiative cooling material **1411**, or it may be shaped in an upward-sloping manner, as shown by radiative cooling material **1421**. Though components besides perimeter walls **1408** and base material **1407** are omitted from the depictions in the inset of FIG. **14**, this is merely for clarity and ease of illustration; it will be understood that radiative cooling material **1411** or radiative cooling material **1421** may be integrated with the other components of structure **1400**, as shown.

[0048] As illustrated in FIG. **14** and as annotated in the inset of FIG. **14**, perimeter walls **1408** may extend above a height of radiative cooling material **1401**, **1411**, or **1421**. Such extended walls may serve as a wind screen and/or as a water collection basin. If structure **1400** includes a substantively flat radiative cooling material (e.g., radiative cooling material **1401**), then a water collection hole (e.g., as annotated in the inset of FIG. **14**) may be included at any suitable location. If structure **1400** includes a non-flat radiative cooling material (e.g., radiative cooling material **1411**, radiative cooling material **1421**, or any other suitable radiative cooling material), then there may be at least one hole arranged near a lowest point of the radiative cooling material and configured to release water (e.g., runoff water) via gravity-driven flow from the collection basin. For example, as shown in the inset of FIG. **14** in connection with radiative cooling material **1411**, the at least one water collection hole may be in perimeter walls **1408**; as also shown in the inset of FIG. **14** in connection with radiative cooling material **1421**, the at least one water collection hole may be in base material **1407**. There may be a corresponding collection system (e.g., as shown and described in connection with FIG. **13**) to catch the runoff water.

[0049] FIG. **15** shows an illustrative structure **1500** including radiative cooling material **1501**, support structure **1508**, and internal barrier **1506**, arranged around roof vent **1504** and above roof **1510**, in accordance with some embodiments of the present disclosure. In some embodiments, structure **1500** is an air cavity configured to feed precooled air into roof vent **1504**. Due to the geometry of support structure **1508**, as shown, and the density differences between cold and warm air, as discussed, structure **1500** exhausts warm air upwards (away from roof vent **1504**) and directs cool air downwards (toward roof vent **1504**). As shown, radiative cooling material **1501** may be arranged over the top surface of support structure **1508**, as well as along at least some of the inner walls (with respect to the air cavity surrounding roof vent **1504**) of support structure **1508**. Based on this arrangement, air within the enclosure surrounded by support structure **1508** is cooled, causing there to be cool air intake at roof vent **1504**. As annotated and as shown by the first type of arrow in FIG. **15**, surrounding air may be drawn into support structure **1508** (e.g., due to convection). This air intake may hit the internal barrier **1506** (e.g., due to the barrier being oriented normal to the primary flow direction of the air intake). As annotated and as shown by the second type of arrow in FIG. **15**, relatively warm air rises from internal barrier **1506** and exhausts out of support structure **1508**. As annotated and as shown by the third type of arrow in FIG. **15**, relatively cool air falls from internal barrier **1506**, collects around roof vent **1504**, and is ultimately drawn into roof vent **1504** as cool air. This relatively cool air may be further cooled (e.g., due to sitting in the air cavity and contacting the radiative cooling material **1501** on the interior walls of the support structure **1508**) before entering roof vent **1504**.

[0050] Support structure **1508** can intake air from multiple sides at an elevated height (with respect to roof vent **1504**). Some of radiative cooling material **1501** is applied over the top of support structure **1508** to cool the air underneath it, which pools at the bottom of support structure **1508**, near roof vent **1504**, due to increased density. The internal barrier **1506** can also break fast winds driving the surrounding air intake, to increase the amount of time that radiative cooling material **1501** has to precool the intake air. The warm air exhaust, as shown, may occur through at least one vent arranged at the top of support structure **1508**, which would also promote regular air circulation

through the enclosure.

[0051] In some embodiments, support structure **1508** may be sealed along the top (e.g., there would be no top vent or opening for warm air exhaust), and warm air may simply reside at the top of support structure **1508** (e.g., away from roof vent **1504**) and/or may flow out of support structure **1508** through the surrounding air intake channel.

[0052] FIG. **16** shows cross section and isometric views of an illustrative radiative cooling system **1600**, including a radiative cooling material arranged on a roof and coupled to a roof vent, in accordance with some embodiments of the present disclosure. Consistent with other embodiments of this disclosure, the roof has a roof vent that intakes air to the building from an air intake pathway (e.g., an air cavity). A support structure is configured to create the air cavity and includes a shim base that is glued to the roof. A radiative cooling material is applied to the top of the structure such that the radiative cooling material cools air of the air cavity and therefore cools air of the air intake pathway. As shown in the isometric view, the radiative cooling material may cover a large portion of the rooftop to maximize a cooling power of the radiative cooling system **1600**.

[0053] FIG. **17** shows a flowchart of an illustrative method for air precooling, in accordance with some embodiments of the present disclosure. At step **1701**, method **1700** includes arranging a radiative cooling material (e.g., any of the radiative cooling materials shown or described in connection with FIGS. **2-16**) upstream of an air intake system (e.g., any roof vent shown in this disclosure, or any other suitable air intake system). At step **1702**, method **1700** includes causing air flowing into the air intake system to be precooled based on the air being cooled by the radiative cooling material. The air may be cooled by the radiative cooling material due to flowing along, over, around, or through the radiative cooling material, or through any other interaction with the radiative cooling material.

[0054] In the following, additional details are provided for various air precooling approaches, consistent with some embodiments of the present disclosure. For example, the following descriptions may apply to any of the radiative cooling approaches shown and described in connection with any one or more of FIGS. **2-17**.

[0055] In some embodiments, a radiative cooling material is used as part of a heat exchanger that exchanges heat between ambient air and any other suitable material (e.g., air of an air cavity, the internal space of a building, or a cooling fluid).

[0056] In some embodiments, a radiative cooling material may be positioned on a roof such that the surface of the radiative cooling material has a relatively unobstructed line-of-sight to the overhead sky. In some embodiments, the air that is cooled by the radiative cooling material may then be used in condensers, in fluid coolers, for building cooling, or for any other suitable purpose. By cooling the intake air to below the ambient temperature, greater heat rejection can be achieved by condensers and fluid coolers, less energy is needed to cool buildings, and buildings can be made more tolerable with less air conditioning.

[0057] In some embodiments, a structure that supports a radiative cooling material is configured (e.g., via a suitable geometric design) to impart a minimal pressure drop to air flowing through it or over it. The structure may also provide sufficient surface area to cool the flowing air. As a result, the temperature of the cooled air may be reduced to below the ambient temperature that is measured on the roof of the building.

[0058] In some embodiments, the nominal heat rejection by a radiative cooling material, when used to cool air to sub-ambient temperatures, may be up to 100 W/m² during the day, and this heat rejection may be greater than 100 W/m² during the night.

[0059] In some embodiments, convective heat transfer due to wind or other fluid flow may warm up an outward-facing surface of a radiative cooling material. Accordingly, various screens, perimeters, baffles, and/or other techniques as provided in some embodiments of this disclosure may be incorporated with a radiative cooling material to manage the convective heating effect and improve the cooling performance.

[0060] In some embodiments, the radiative cooling material is on a roof and exposed to the sky, creating the potential to become dirty. Radiative cooling materials with dirty surfaces may not provide as much heat rejection as they do with clean surfaces. Accordingly, various screens, perimeters, baffles, and/or other techniques as provided in some embodiments of this disclosure may be incorporated with a radiative cooling material to prevent dirt accumulation and/or to facilitate cleaning.

[0061] In some embodiments, including when this radiative cooling device or structure is used as a precooler for air entering buildings, the radiative cooling material can either be directly applied on the roof around the intake vent (e.g., without applying the material over a support structure), or the material can be applied over a structure. In some embodiments, the material and/or the support structure may direct the cooled air to a vent, condenser or other air conveyance system.

[0062] In some embodiments, a radiative cooling structure can be applied to many different types of air intake vents including louver air intake vents, ventilator, box roof vents, roof fans, hooded ventilators, any other suitable air intake equipment, or any combination thereof. Additionally, a radiative cooling structure can be used with fluid coolers, evaporative condensers, air cooled condensers, any other suitable cooling equipment, or any combination thereof.

[0063] In some embodiments, including when a radiative cooling material is directly applied to the roof, a barrier may be created around the radiative cooling material to reduce mixing the colder air on the roof surface from the freestream ambient air above the roof. The barrier may also direct cold air (e.g., due to it having a higher density than warmer air), into the intake vents, fans or other air conveyance systems while rejecting warm air from entering these same systems.

[0064] In some embodiments, to manage air flow proximal to a radiative cooling material, air can be pulled over, under, and/or around a radiative cooling material to maximize cooling by maximizing the amount of time during which the air interfaces with the radiative cooling material. In some embodiments, the air can be pulled through a radiative cooling material that has holes, perforations or slits to maximize the amount of heat rejection that the cooling surface can provide to the air. These holes, perforations, or slits may generally be referred to as multiple openings in the radiative cooling material.

[0065] In some embodiments, structures supporting the radiative cooling material can be installed using mechanical fasteners directly to the roof of a building, or ballasted using materials that can also act as thermal energy storage materials. For example, the structure supporting the radiative cooling material can be designed to have a thickness such that it will be cooled by the radiative cooling material at night and during the early morning (e.g., when the sun is not out or when it is low on the horizon), and this cooled support material may slowly release the heat later in the day when the ambient temperature is warmer, thereby improving a cooling power of the system by providing thermal energy storage.

[0066] In some embodiments, the structure supporting a radiative cooling material can be installed with thermal insulation to improve cooling of an air cavity (at least partially) enclosed by the radiative cooling material, or at the surface of the radiative cooling material. In some embodiments, such insulation would be improve the cooling power when significant heat is conducted through a roof, which could otherwise hinder the cooling performance of the radiative cooling material.

[0067] In some embodiments, further increased cooling power and/or cooling efficiency can be achieved by using wind screens that are built into or around a structure supporting a radiative cooling material to minimize convective heat transfer to the surface of the radiative cooling material. In some embodiments, wind screens or other techniques may cause a layer of cold air to accumulate (e.g., even in the absence of any sealing or air cavity, due to density) at the skyward facing surface of the radiative cooling material, and this layer of cold air may be directly coupled to an air vent or other air conveyance system. In some embodiments, wind screens or other devices may also be used to force air to flow through or around the radiative cooling surface.

[0068] In some embodiments, any wind screen provided in some embodiments of this disclosure

may be arranged no less than 3" above the surface of the radiative cooling material to provide an air gap (e.g., forming cavity of cooled air) between the screen and the surface.

[0069] In some embodiments, a structure supporting a radiative cooling material can be glued or otherwise adhered to a roof (or another suitable surface) via a plastic curb or shim. These support structures may be configured for temporarily use and/or releasable affixation to the roof.

[0070] In some embodiments, a structure supporting a radiative cooling material has corrugations on it. The corrugations are configured to cause runoff of surface moisture (e.g., when it rains). That surface moisture runoff can carry debris (e.g., dust dirt, or any other matter on the surface of the radiative cooling material) from the radiative cooling material, such that the corrugations help keep the radiative cooling material clean. In some embodiments, low points in the corrugations can have slits, holes or other drainage canals to remove and/or collect rainwater, debris, and other runoff.

[0071] In some embodiments, during the mornings and evenings, a radiative cooling material can cool air to below the dew point, resulting in moisture condensing on the surface of the radiative cooling material. This moisture can be collected and harvested for drinking water by angling the surface and having water collection vessels at a corresponding end of the angled structure.

[0072] In some embodiments, a radiative cooling material may have holes, slits, perforations, or other permeability features (e.g., any suitable plurality of openings) such that air can flow through the radiative cooling material. Allowing air to flow through the radiative cooling material may improve the cooling power of the material. In some embodiments, the perforations, slits, holes, or other suitable openings are implemented such that they do not reduce the sky facing surface area of the radiative cooling material.

[0073] In some embodiments, a structure supporting a radiative cooling material is arranged in direct contact with existing air exhaust vents or intake vents (e.g., including intake plenums). The radiative cooling material may be directly contacted to such vents with a rubber gasket, shroud or other substantially airtight liner to minimize (or eliminate) the entry of warm air (e.g., as heated by the roof) into the precooled air cavity or the corresponding air handling system.

[0074] In some embodiments, a structure supporting a radiative cooling material can have a walkway that allows access to the air intake equipment, exhaust vent, fluid cooler and condenser equipment (e.g., for maintenance).

[0075] In some embodiments, there may be a flexible air shield or shroud between the edge of the air vent and the structure that holds a radiative cooling material.

[0076] In some embodiments, a radiative cooling system can itself be fashioned into (or in direct contact with) a roof for a building (e.g., a barn, greenhouse, or data center), such that air is cooled while it is pulled inside the underlying building.

[0077] In some embodiments, for a given rooftop vent, the volume of air flow through the vent will determine a suitable area of radiative cooling material to apply to the roof. For a volumetric air flow Q , the low and high area estimates may be determined by:

$$[00001] A = \frac{\text{AirFlow} \cdot \text{Math.} \cdot \text{Math.} \cdot c_p \cdot \text{Math.} \cdot T}{\text{radiativecoolingheatflux}(f(T))} \cdot \text{Math.} \cdot \text{con}$$

where ρ is the density of the air, $c_{\text{sub.p}}$ is the heat capacity of air, ΔT is the temperature difference between the surface and the entering air, and the radiative cooling heat flux is a function of the temperature difference between the ambient and the surface. Additionally, con is a system constant that is defined by the site conditions and may be used to estimate the area of radiative cooling material needed to cool enough air for this application. That is, a size of the radiative cooling material may be configured according to expected cooling needs and/or required cooling power at a given site.

[0078] In some embodiments, a radiative cooling material is configured to integrate with existing buildings, roofs, condensers and fluid coolers with minimal modification to the roof of a building. Compared to evaporation cooling systems, the corresponding radiative cooling system would not require the use of water.

[0079] In some embodiments, a thermal mass is provided for thermal energy storage of cold temperatures generated by a radiative cooling material. The thermal mass material may have a relatively high specific heat capacity (e.g., as measured in J/kgK) to effectively maintain a relatively low temperature and a high thermal conductivity (e.g., as measured in W/mK) to avoid a significant temperature difference between the air and the radiative cooling material. The thermal mass may be arranged, e.g., on a roof surface to cause a radiative cooling system to maintain a low temperature without further increasing the temperature differential between the radiative cooling material and air temperature. Thermal mass materials with relatively high heat capacity and relatively high R-value may be preferable (e.g., as opposed to high heat capacity and low R-value) when thermally coupled to the surface of the radiative cooling material. Certain nonlimiting thermal mass materials are considered and characterized in the following table:

TABLE-US-00001

TABLE 1 List of illustrative thermal mass materials and relevant properties.			
Thermal Heat Capacity: (J/kg K)	Conductivity: k (W/mK)	Material	C. Radiative cooling Diamond
5090	2000	material attached Graphite (In Plane: 720 2500 Thermal Mass Parallel to layers) (High C. High k)	Aluminum Nitride 760 250 Roof attached Water 4180 0.6 Thermal Mass Concrete 880 1.7 (High C. low k)
Wood 2300	0.13	Aerogels 700	0.004 Air 1005 0.026

[0080] In some embodiments, including any one or more of those described above, the radiative cooling material, structures to which the radiative cooling material is applied, and other materials that form parts of a radiative cooling system may be attached to a roof over which the radiative cooling system is applied with conventional rooftop supports, ballast or other fasteners.

[0081] In some embodiments, including any one or more of those described above, an air cavity holding air that is cooled by a radiative cooling material may be sealed off, such that cool air cannot escape from an air intake path and such that warm air outside the air cavity cannot mix with the cool air. That cooled air may be referred to as precooled air if, after having been cooled, it flows into any air intake channel or system.

[0082] In some embodiments, a radiative cooling material has a finned surface, and the finned surface is arranged opposite to the surface of the radiative cooling material that is configured to face the sky (e.g., and thereby generate cooling). In some embodiments, an air cavity or other air intake channel may be configured such that prior to entering an air intake system, the air flows across the finned surface.

[0083] It will be understood that in this disclosure, an air intake may be any device, structure, or suitable combination thereof, that receives air (e.g., a roof vent, any other vent, any duct, any air piping, any air channel, any other passage into the interior of a building, or any combination thereof). As used herein, an air intake system may include any suitable air intake (e.g., any roof vent showed herein), and it may optionally further include any number of additional devices, materials, structures, or any combination thereof, e.g., for routing, moving, handling, conditioning, precooling, or otherwise managing the intake air. For example, an air intake system may include an air intake (e.g., any roof vent shown in this disclosure), it may further include an air cavity arranged upstream of the air intake (e.g., as shown at least in FIGS. 2-11, and 14-15), and it may further include an air-cooled condenser or other cooling equipment arranged downstream of the air intake. The air intake system may include a radiative cooling material (e.g., coupled to the air intake and/or to the air cavity) that is configured to cool the air prior to the air entering the air intake (i.e., to precool the air). That precooled air may be provided to the interior of a building, to additional air handling equipment, to any other suitable structure, or to any combination thereof.

[0084] It will be understood that in this disclosure, when describing the arrangement of a radiative cooling structure, “above”, “over”, and related orientational language is used with respect to a typical arrangement of a roof being atop a building, and the radiative cooling structure being atop the roof (e.g., as viewed from ground level).

[0085] The processes described above are intended to be illustrative and not limiting. One skilled in the art would appreciate that the steps of the processes described herein may be omitted, modified,

combined and/or rearranged, and any additional steps may be performed without departing from the scope of the invention.

[0086] The foregoing is merely illustrative of the principles of this disclosure, and various modifications may be made by those skilled in the art without departing from the scope of this disclosure. The above-described embodiments are presented for purposes of illustration and not of limitation. The present disclosure also can take many forms other than those explicitly described herein. Accordingly, it is emphasized that this disclosure is not limited to the explicitly disclosed methods, systems, and apparatuses, but is intended to include variations thereto and modifications thereof, which are within the spirit of the following claims.

Claims

1. A system comprising: at least one radiative cooling material; and at least one air intake, wherein: the at least one radiative cooling material is arranged upstream of the at least one air intake, and air moves from the at least one radiative cooling material to the at least one air intake, such that the air is cooled using the at least one radiative cooling material prior to entering the at least one air intake.
2. The system of claim 1, wherein: the at least one radiative cooling material and the at least one air intake are arranged on a roof of a building; and the at least one air intake provides the cooled air to inside the building.
3. The system of claim 1, wherein the at least one air intake comprises at least one of an air vent, air conditioner, or air cooled condenser.
4. The system of claim 1, wherein: the at least one air intake comprises an outer surface and a vent, the outer surface being arranged upstream of the vent, the at least one radiative cooling material is arranged on the outer surface, and the air being cooled air using the at least one radiative cooling material prior to entering the at least one air intake comprises the air being cooled by the outer surface prior to entering the vent.
5. The system of claim 1, further comprising an air cavity arranged upstream of the at least one air intake, wherein: the at least one radiative cooling material is arranged on at least one surface enclosing the air cavity, and the cooled air entering the at least one air intake is provided by the air cavity.
6. The system of claim 5, wherein the air cavity is formed by a volume enclosed on a first side by a roof, enclosed on a second side by the at least one air intake, and enclosed on at least one other side by the at least one surface on which the at least one radiative cooling material is arranged.
7. The system of claim 5, wherein the at least one air intake is coupled to an intake plenum, wherein the intake plenum forms a seal between the air cavity and the at least one air intake.
8. The system of claim 1, wherein the at least one air intake is on a roof, and the at least one radiative cooling material is arranged above the roof, such that there is an air gap between the roof and the at least one radiative cooling material.
9. The system of claim 1, further comprising a wind screen arranged over the at least one radiative cooling material, wherein the wind screen reduces convective heating of the at least one radiative cooling material.
10. The system of claim 1, further comprising a support structure, wherein: the at least one radiative cooling material is applied to the support structure; and the support structure directs the cooled air to enter the at least one air intake.
11. The system of claim 10, wherein the support structure comprises corrugations configured to prevent liquid or solid matter from accumulating on the at least one radiative cooling material.
12. The system of claim 1, wherein the at least one radiative cooling material comprises a plurality of openings through which the air flows before entering the at least one air intake, wherein the air flow through the plurality of openings causes the air to be cooled.

- 13.** The system of claim 1, further comprising a thermal mass that is thermally coupled to the at least one radiative cooling material.
- 14.** The system of claim 1, further comprising a moisture collector configured to collect moisture that accumulates on the at least one radiative cooling material.
- 15.** A method comprising: arranging at least one radiative cooling material upstream of at least one air intake; and cooling air that moves from the at least one radiative cooling material to the at least one air intake using the at least one radiative cooling material, such that the air is cooled prior to entering the at least one air intake.
- 16.** The method of claim 15, wherein: the at least one radiative cooling material and the at least one air intake are arranged on a roof of a building; and the at least one air intake provides the cooled air to inside the building.
- 17.** The method of claim 15, wherein: the at least one air intake comprises an outer surface and a vent, the outer surface being arranged upstream of the vent, arranging the at least one radiative cooling material upstream of the of the at least one air intake comprises arranging the at least one radiative cooling material on the outer surface, and cooling the air such that the air is cooled prior to entering the at least one air intake comprises using the outer surface to cool the air such that the air is cooled prior to entering the vent.
- 18.** The method of claim 15, further comprising arranging an air cavity upstream of the at least one air intake, wherein: the at least one radiative cooling material is arranged on at least one surface enclosing the air cavity, and the cooled air entering the at least one air intake is provided by the air cavity.
- 19.** The method of claim 15, wherein the at least one radiative cooling material comprises a plurality of openings through which the air flows before entering into the at least one air intake, wherein the air flow through the plurality of openings causes the air to be cooled.
- 20.** A system comprising: at least one radiative cooling material; at least one air intake, wherein: the at least one radiative cooling material is arranged upstream of the at least one air intake, and air moves from the at least one radiative cooling material to the at least one air intake, such that the air is cooled using the at least one radiative cooling material prior to entering the at least one air intake; and air handling equipment configured to: receive the cooled air from the at least one air intake, further cool the received cooled air, and provide the further cooled air to a cooling system.
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